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IT4010 Research Project – 2026 Jul

Topic Assessment Form

V2.2

Project ID: J26-CS-318

1. Topic (12 words max)

Protocol-aware ML framework for intelligent Modbus attack detection and incident analysis

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Cyber Security (CS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Nature and Gravity of the Problem

Industrial Control Systems (ICS) and Supervisory Control and Data Acquisition (SCADA) environments for the backbone of modern critical infrastructure such as power generation, water treatment, manufacturing and oil gas pipelines [1], [4]. Large proportions of these systems communicate using Modbus/TCP protocol, which was designed decades ago for reliability and simplicity rather than for security [8], [9]. Modbus contains no native authentication, no encryption and no integrity checking, which means that any device able to reach the network can read, process data, issue control commands or impersonate legitimate masters and slaves [11], [13].

As ICS networks are increasingly connected to corporate networks and the internet to enable remote monitoring, the attack surface has expanded dramatically [4], [11]. Adversaries can now launch Denial-of-Service (DoS) attacks that flood Programmable Logic Controllers (PLCs) with malformed or high-frequency requests, reconnaissance scans that quietly map registers and devices, command and response injection that forges process commands, and replay attacks that re-transmit previously captured legitimate packets [13]. Conventional Intrusion Detection Systems (IDS) deployed in these networks rely mainly on signature-based or static rule-based methods and on generic network-flow features that ignore the semantics of the Modbus protocol [1,], [6], [15]. As a result they fail to detect novel, low-volume or protocol-valid attacks, and produce high false-positive rates that overwhelm operators [10], [15].

This research therefore investigates a unified, protocol-aware machine learning framework that detects and explains four distinct classes of Modbus attack; DoS, reconnaissance, command/response injection and replay. Each through a methodology tailored to the behavioral

Signature of that attack, integrated into a single intelligent detection and incident-analysis pipeline.

Impact of the problem

- **Operational and safety –** undetected attacks on PLCs and field devices can cause incorrect actuator behaviour, equipment damage, process disruption and even physical safety hazards for personal and the public [4], [13].
- **Technical –** existing IDS depend on generic flow features and signatures, cannot model protocol semantics or temporal behaviour, and degrade against zero-day or protocol-valid attacks [1], [6], [15].
- **Economic –** downtime in critical infrastructure, recovery costs and regulatory penalties are extremely high, while specialised commercial OT security appliances are costly and difficult to adapt [4], [11].
- **Privacy and trust –** manipulation or exfiltration of process data undermines the integrity and trustworthiness of national infrastructure and the data it generates [7], [11].

FoC research cluster and industry verticals

- **Research cluster –** cyber security and network engineering / Software Systems and Technologies (SST) focusing on protocol aware, machine-learning based intrusion detection and incident-analysis software for operational technology networks.
- **Industry verticals –** Critical Infrastructure Protection, Operational Technology (OT) / ICS-SCADA Security, Energy and Utilities, and Smart Manufacturing (Industry 4.0).

Sustainable Development Goals (SDGs) attained

- **SDG 9 (Industry, Innovation and Infrastructure) –** strengthens the resilience and reliability of industrial infrastructure by detecting and explaining cyber-attacks against Modbus-based control systems.
- **SDG 7 (Affordable and Clean Energy) –** Helps protect energy generation and distribution systems, frequent Modbus deployments from disruption, supporting reliable and continuous energy supply.
- **SDG 11 (Sustainable Cities and Communities) –** contributes to safe and resilient water, transport and utility services that depend on secure industrial control systems.

6. Description of Existing Research and Systems (300 words max)

Summary of existing products/systems

Most operational deployments protect Modbus networks using perimeter firewalls, industrial demilitarized zones and signature-based intrusion detection sensors such as Snort with hand-written ICS rule sets [15]. Commercial OT-security platforms provide deep-packet inspection and asset discovery for Modbus and similar protocols, but they are expensive, largely rule-driven and tuned to known attack signatures [15]. Research testbeds such as the Mississippi State University gas-pipeline and water-storage datasets [2], [8], recent Modbus/TCO attack-emulation testbeds [13], [18] have been used to generate labelled normal and attack traffic covering reconnaissance, injection, DoS and replay scenarios.

Summary of existing research

Machine learning and deep learning have become dominant research direction for ICS intrusion detection [1], [4]. Supervised classifiers such as Decision Trees, Random Forest, SVM, KNN and XGBoost have been applied to Modbus traffic, with Random Forest and gradient boosting models reaching very high reported accuracy on benchmark datasets [5], [6]. Deep models including DNN, CNN and hybrid GooLeNet-LSTM architecture have been used to capture spatial and temporal patterns in ICS communication for mutli-class attack detection [3]. Unsupervised approaches particularly autoencoders that learn a model of normal traffic and flag high reconstruction error have proven effective for anomaly and zero-day detection in SCADA, DNP3 networks, including function-code based autoencoder IDS designs [9], [13]. Protocol-specific work has demonstrated replay-attack detection using timestamps and sequence numbers [10] and explainable-AI methods that take Modbus IDS decisions interpretable [6], [7].

Gaps/limitations identified

- Most IDS use generic flow features (packet size, bytes/sec, TCP flags) and ignore Modbus protocol semantics such as function codes and register access patterns [1], [6].
- Signature and rule-based system cannot detect novel, low-volume or protocol-valid attacks and generate high false-positive rates [15].
- Reconnaissance scans rarely change traffic volume, so packet-level detectors miss the subtle temporal behavior that distinguishes them [3], [18].
- Injection attacks are frequently inspected per-packet, ignoring whether a command is valid within its surrounding communication context [9], [13].
- Replay attacks reuse genuinely valid packets, so content-based detection alone fails without transaction-consistency analysis [10].
- Few works integrate detection of multiple attack classes into one framework with unified incident analysis and explanation [4], [7].

Hence there is a clear need for a single, cost-effective, protocol-aware machine-learning framework that combines attack-specific detection methodologies with unified incident analysis for Modbus-based industrial control systems.

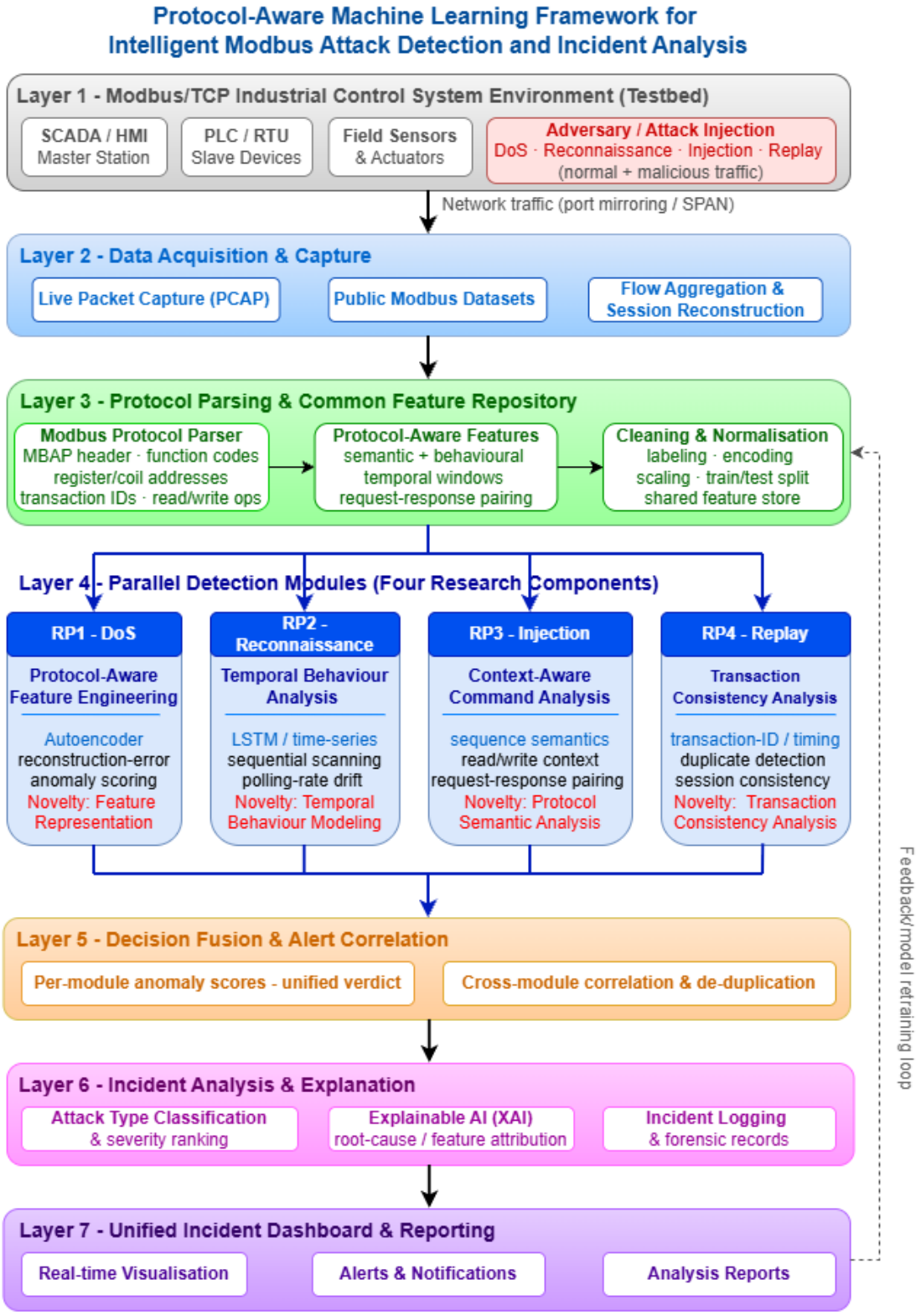
7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

Addressing the identified research gaps

The proposed solution is a Protocol-aware machine learning framework that captures live and recorded Modbus/TCP traffic, parses it into a shared, protocol-aware repository, and then routes those features to four parallel detection modules, one per attack class, before fusing their outputs into a single incident-analysis and reporting layer. Rather than treating every attack with one generic model, each research components apply the methodology best matched to the behavioral signature of its target attack.

- **RP1 – DoS via Protocol-aware feature engineering:** engineers Modbus-specific features (function codes, register/coil addresses, transaction IDs, read/write behaviour) and uses an autoencoder to flag flooding and resource-exhaustion anomalies that generic flow feature miss.
- **RP2 – Reconnaissance via temporal behavior analysis:** models communication patterns over time using time-series/LSTM techniques to reveal sequential register scanning and abnormal polling frequency that produce no obvious volume spike.
- **RP3 – Command/response injection via context-aware command analysis:** analyses command relationships and protocol semantics. For example, an unexpected write after read, invalid function-code usage, or abnormal request-response pairing instead of inspecting packet in isolation.
- **RP4 – Replay via transaction consistency analysis:** examines transaction IDs, response timing, duplicate requests and session consistency to separate replayed but valid packets from genuine communication.

A common feature repository removes duplicated preprocessing, lets the four modules share a consistent view of the traffic, and supports cross-module correlation. A decision-fusion layer combines per-module anomaly scores into a unified verdict, and an incident-analysis layer adds attack-type classification, severity ranking and explainable-AI attribution so that operators understand why an alert was raised. Because the framework relies on open datasets, an inexpensive testbed and software models rather than proprietary appliances, it offers an affordable and adaptable alternative for securing Modbus ICS networks.


Protocol-Aware Machine Learning Framework for
Intelligent Modbus Attack Detection and Incident Analysis
Layer 1 - Modbus/TCP Industrial Control System Environment (Testbed)
SCADA / HMI
Master Station
PLC / RTU
Slave Devices
Field Sensors
& Actuators
Adversary / Attack Injection
DoS · Reconnaissance · Injection · Replay
(normal + malicious traffic)
Network traffic (port mirroring / SPAN)
Layer 2 - Data Acquisition & Capture
Live Packet Capture (PCAP)
Public Modbus Datasets
Flow Aggregation &
Session Reconstruction
Layer 3 - Protocol Parsing & Common Feature Repository
Modbus Protocol Parser
MBAP header · function codes
register/coil addresses
transaction IDs · read/write ops
Protocol-Aware Features
semantic + behavioural
temporal windows
request-response pairing
Cleaning & Normalisation
labeling · encoding
scaling · train/test split
shared feature store
Layer 4 - Parallel Detection Modules (Four Research Components)
RP1 - DoS
Protocol-Aware
Feature Engineering
Autoencoder
reconstruction-error
anomaly scoring
Novelty: Feature
Representation
RP2 -
Reconnaissance
Temporal Behaviour
Analysis
LSTM / time-series
sequential scanning
polling-rate drift
Novelty: Temporal
Behaviour Modeling
RP3 - Injection
Context-Aware
Command Analysis
sequence semantics
read/write context
request-response pairing
Novelty: Protocol
Semantic Analysis
RP4 - Replay
Transaction
Consistency Analysis
transaction-ID / timing
duplicate detection
session consistency
Novelty: Transaction
Consistency Analysis
Layer 5 - Decision Fusion & Alert Correlation
Per-module anomaly scores - unified verdict
Cross-module correlation & de-duplication
Layer 6 - Incident Analysis & Explanation
Attack Type Classification
& severity ranking
Explainable AI (XAI)
root-cause / feature attribution
Incident Logging
& forensic records
Layer 7 - Unified Incident Dashboard & Reporting
Real-time Visualisation
Alerts & Notifications
Analysis Reports
Feedback/model retraining loop

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Gaining specialized domain expertise

The team will build domain expertise in industrial control systems and operational-technology security by consulting academic supervisors, cyber-security practitioners and where possible, OT/ICS engineers who operate Modbus-based equipment. The group will study how Modbus/TCP communication works in practice, the master-slave request/response model, function codes, register and coil addressing and the MBAP header and how each target attack manifests at the protocol level [8], [11], [13]. Members will review the structure and behavior of SCADA/HMI systems, PLCs and field devices so that modelling of normal versus malicious behavior is grounded in realistic operational patterns. Regular discussion with supervisors and review of standards and case studies on ICS incidents will deepen the team's understanding of attacker techniques, defensive controls and the safety implications of false negatives and false positives in critical infrastructure.

Gaining technical knowledge and skills

The technical skills required for the project span network traffic analysis, protocol parsing, feature engineering and machine/deep learning. Team members will acquire these through online courses, academic literature, technical documentation and hands-on experiment. Specific competencies include packet capture and dissection (e.g- with Wireshark/tshark and Scapy), building and labelling datasets, and engineering protocol-aware features from Modbus traffic. On the modelling side, members will learn to design and evaluate autoencoders for anomaly detection, LSTM and other time-series models for temporal behavior, sequence and context modelling techniques for command analysis, and consistency-checking methods for transaction analysis [3], [9], [12]. The team will also develop skills in building the integration layer and incident dashboard, including data visualization, model service and explainable AI techniques for interpreting detections [6], [7]. Standard evaluation methodology with accuracy, precision, recall, F1 score, false positive rate and ROC-AUC will be used to compare models against baselines [5], [6], [10].

Data collection

The project will use both primary and secondary data. Primary data will be generate from a controlled Modbus/TCP testbed using physical or simulated PLCs and a Modbus master, supported by simulation frameworks such as ICSSIM and Modbus attack-emulation tools in which the four target attacks (DoS, reconnaissance, command/response injection and replay) are executed to capture labelled normal and malicious traffic [13], [18]. Secondary data will be drawn from established public ICS/SCADA datasets, including the Mississippi State University gas-pipeline and water-storage datasets and recent Modbus-specific anomaly detection datasets, to benchmark the models and improve generalization [2], [8], [17]. Collected traffic will be parsed, cleaned, labelled and split into training and testing sets within the common feature repository, and will be the basis for training and validating every detection module.

Ethical clearance

Because the research uses network traffic from controlled testbeds and publicly available datasets rather than personal or medical data, the ethical risk is low. No human subject data is collected. All traffic is generated in an isolated laboratory environment so that no live or third-party infrastructure is affected. Any third-party datasets will be used strictly in accordance with their licenses and only for academic research. The project will follow responsible disclosure principles for any vulnerabilities observed and will adhere to the university's research ethics guidelines, ensuring that the work is conducted safely, legally and for defensive purposes only.

9. Objectives and Novelty

Main Objective To design and develop a unified, protocol-aware machine learning framework that detects correlates and explains multiple classes of Modbus attacks: Denial-of-Service, reconnaissance, command/response injection and replay in industrial control systems, providing real-time detection and incident analysis through a single intelligent pipeline.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT23219984 Rathnayake H.M.S.R	DoS Attack detection using protocol-aware feature engineering.	• Build a Modbus protocol parser and extract protocol-aware features (function codes, register addresses, transaction IDs, read/write behavior. • Generate and label DoS attack traffic on the testbed and combine it with public datasets. • Design and train an autoencoder to model normal Modbus behavior and flag DoS anomalies via reconstruction error. • Compare protocol-aware features against generic flow features.	Introduces protocol-aware feature engineering for Modbus DoS detection. Unlike conventional IDS that rely on generic network-flow features, this component evaluates whether Modbus specific semantic features improve autoencoder-based detection of flooding and resource exhaustion attacks.

		• Evaluate using accuracy. Precision, recall, F1 and false positive rate.	
IT23269002 Charith H	Reconnaissance attack detection using temporal behavior analysis	• Capture and label reconnaissance/scanning traffic (sequential register scanning, polling-rate changes). • Construct time-windowed and sequence features representing communication behavior over time. • Design and train time-series/LSTM models to detect subtle temporal patterns. • Validate that temporal modelling outperforms packet-level analysis. • Evaluate detection performance and latency.	Models communication patters over time rather than analyzing packets independently. Because reconnaissance rarely changes traffic volume, this component detects scanning through temporal behavior, sequential register access and abnormal polling frequence that volume-based detectors miss.
IT23150270 Rathnayaka K.H	Command and response injection detection using context-aware command analysis	• Extract command sequence and request response features from Modbus traffic. • Define legitimate protocol-semantic context (valid function code use, read/write ordering, request-response pairing).	Introduces context-aware analysis of Modbus command sequences using protocol semantics instead of only statistical traffic features. It detects injection by reasoning about command relationships and request-response context rather than inspecting packets in isolation.

		• Design models that judge whether the command is valid within its surrounding context. • Detect injected commands such as unexpected writes, invalid function codes and illegal register modifications. • Evaluate against per-packet baselines.	
IT23284616 Ilankoon I.M.D.S	Replay attack detection using transaction consistency analysis	• Capture and label replay attack traffic on the testbed. • Extract transaction-level features such as transaction IDs, response timing, duplicate requests, session consistency. • Design a transaction consistency model to distinguish replayed packets from legitimate traffic. • Integrate detection results into the fusion and incident-analysis layer. • Evaluate detection accuracy and false-positive rate.	Detects inconsistencies across transactions rather than withing individual packets. Because replayed packets are individually valid, this component uses communication state information such as transaction IDs, timing and session consistency. Instead of packet content alone.

10. <u>This section is to be completed by the Supervisor and the Co-supervisor of the Project.</u>

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Data required for the project is available.

Yes		No	

f) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor*				
Co-Supervisor*				

References

References

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